Medical Imaging & Digital Twin Specification	Specify MRI preprocessing, segmentation, geometry, tumor tracking and 3D visualization	Critical	
06	AI & Multimodal Intelligence Specification	Define imaging AI, clinical NLP, temporal models, knowledge retrieval and evidence-linked interpretation	Critical
07	System Architecture & Infrastructure Blueprint	Define backend, frontend, databases, API boundaries, storage, cloud inference and deployment	Critical
08	Database & API Contracts	Specify database schemas, endpoints, request/response structures and state updates	Critical
09	UI/UX & Interactive Demonstration Specification	Define screens, navigation, 3D interactions, timeline, charts and visual design	Critical
10	AI Safety, Privacy & Scientific Integrity	Set rules for synthetic data disclosure, uncertainty, provenance and clinical claim boundaries	Critical
11	Testing & Scientific Evaluation Plan	Define automated tests, real-data validation, metrics, baselines and failure criteria	Critical
12	Implementation Roadmap & Finalist Demo Plan	Convert requirements into prioritized coding tasks, milestones and the final presentation sequence	Critical

3. Preparations beyond documentation

We should prepare five practical resources before starting development.

A. Dataset readiness

- Identify I-SPY2, I-SPY1, Duke and BreastDCEDL dataset access requirements.

- Download a small sample of breast MRI images and associated metadata.
- Verify dataset licensing and patient de-identification.
- Determine which datasets contain real repeat visits.
- Inspect available segmentation masks and clinical variables.
- Produce a dataset quality and feasibility report.

B. Development environment

- Create the GitHub repository and project directory structure.
- Set up Python, FastAPI, Next.js and the selected database.
- Configure Meta Model API access and credentials.
- Select cloud infrastructure for GPU-dependent imaging operations.
- Prepare local development to operate within the laptop's 8 GB RAM constraint.
- Establish automated checks, tests and CI.

C. AI agent instructions

- AGENTS.md — project-wide coding rules and boundaries.
- ARCHITECTURE.md — service ownership and integration rules.
- TASKS.md — implementation backlog and dependency order.
- DATA_POLICY.md — restrictions on clinical data and simulated outputs.
- DEFINITION_OF_DONE.md — tests and evidence required before marking tasks complete.

D. Research and evaluation assets

- Define five curated simulation cases.
- Prepare sample patient records and expected outputs.
- Define train/validation/test splits by original patient.
- Create reference measurements and segmentation benchmarks.
- Prepare an evaluation dataset independent of the synthetic simulation generator.

E. Competition submission resources

- Confirm accepted track and official eligibility.
- Review the official hackathon rules when published.
- Prepare a demonstration storyboard.
- Plan screenshots, a short demo video and the project README.
- Prepare a technical architecture diagram and limitations statement.

Conduct one consolidated pre-development architecture and specification audit.

This should produce a short report identifying contradictions, unresolved dependencies, missing implementation details and release-blocking risks.

We should not rewrite all 12 documents. We should reconcile their requirements, establish the authoritative contracts, and then let the coding agents begin implementation against the approved baseline.

CancerOS — Master Engineering Prompt Inventory

CODING AGENT EXECUTION PLAN · VERSION 1.0 · HACKATHON PROTOTYPE

Objective: Define the complete inventory of engineering prompts required to take CancerOS from its 12 drafted specifications to a working, tested, deployed, and demonstrable breast cancer digital twin.

This is a prompt planning document, not the actual coding prompts. Each item represents a separately executable engineering assignment that can later be expanded into detailed instructions for Codex or another coding agent.

I recommend 104 engineering prompts, organized into 15 workstreams. Not all are equally important: the P0 assignments establish the functional prototype, while P1 assignments improve technical quality, research capability and presentation.

104

Engineering prompts

15

Workstreams

12

Source specifications

Planning inventory; prompt counts are proposed task boundaries, not estimated development time. The twelve specifications remain the governing source of requirements.

Phase 1 — System governance and repository foundations

These are the highest-leverage prompts. They establish how the coding agent must interpret the specifications, prevent uncontrolled architecture changes, and evaluate its own work.

Workstream 01 — System-wide governance and agent guidelines

8 prompts

ID	Engineering prompt to create	Priority
COS-001	Master system-wide coding-agent instructions and engineering authority hierarchy	P0
COS-002	Twelve-document specification audit and cross-document contradiction detection	P0
COS-003	Requirements traceability matrix and acceptance-criteria mapping	P0
COS-004	Engineering coding standards, naming conventions and repository boundaries	P0
COS-005	Clinical data integrity, AI safety and synthetic provenance coding rules	P0
COS-006	Coding-agent task execution, checkpoint ledger and change-control rules	P0
COS-007	Definition of Done, automated verification and evidence requirements	P0
COS-008	System-wide integration and architectural drift prevention guidelines	P0

Expected deliverables: AGENTS.md, engineering governance rules, an audit report, traceability matrix, task ledger, and definition-of-done checklist.

Workstream 02 — Repository and development environment

7 prompts

ID	Engineering prompt to create	Priority
COS-009	Initialize monorepo or coordinated application repository structure	P0
COS-010	Configure Python backend and dependency management	P0
COS-011	Configure Next.js frontend, TypeScript and design foundations	P0
COS-012	Configure database, local object storage and migrations	P0
COS-013	Configure development containers or lightweight local service startup	P0
COS-014	Configure environment variables, secrets and provider credentials	P0
COS-015	Implement CI pipelines, linting, static checks and test runners	P0

Expected deliverable: A reproducible development environment capable of starting the frontend, backend, persistence services and selected processing workers.

Phase 2 — Data and scientific foundations

Workstream 03 — Clinical data, datasets and provenance

9 prompts

ID	Engineering prompt to create	Priority
COS-016	Implement dataset source registry, licenses and attribution metadata	P0
COS-017	Implement pseudonymous patient identities and source-patient mappings	P0
COS-018	Implement canonical clinical data schemas and controlled terminology	P0
COS-019	Implement clinical visit, treatment and observation ingestion	P0

COS-020	Implement laboratory, vital-sign and symptom validation	P0
COS-021	Implement dataset-specific metadata adapters for approved cohorts	P0
COS-022	Implement provenance lineage, versioning and source immutability	P0
COS-023	Implement data completeness assessment, quarantine and quality reports	P0
COS-024	Implement duplicate detection and patient-level train/validation/test partitioning	P0

Expected deliverable: A source-aware clinical data layer that keeps actual patient observations separate from synthetic records and prevents patient-level data leakage.

Workstream 04 — Medical imaging ingestion and preprocessing

8 prompts

ID	Engineering prompt to create	Priority
COS-025	DICOM and NIfTI ingestion with study/series association	P0
COS-026	Imaging geometry, orientation, spacing and physical-coordinate validation	P0
COS-027	MRI preprocessing, intensity normalization and sequence management	P0
COS-028	Image/mask spatial alignment and resampling integrity	P0
COS-029	Reference segmentation mask import and annotation classification	P0
COS-030	Optional nnU-Net/MONAI automated tumor segmentation integration	P1
COS-031	Segmentation quality control and reference-mask evaluation	P0
COS-032	Imaging processing jobs, artifact storage and failure recovery	P0

Expected deliverable: A reliable pathway from authentic breast MRI and suitable tumor annotations to validated volumetric image assets.

Workstream 05 — Tumor geometry and anatomical digital twin

8 prompts

ID	Engineering prompt to create	Priority
COS-033	Physical tumor measurements and volume calculation engine	P0
COS-034	Segmentation-to-3D-mesh reconstruction pipeline	P0
COS-035	Mesh quality validation, simplification and coordinate preservation	P0
COS-036	Breast-region anatomical context and illustrative geometry labeling	P0
COS-037	Real longitudinal study alignment and registration quality checks	P0
COS-038	Lesion identity matching and temporal measurement tracking	P0
COS-039	Time-indexed digital twin state persistence and immutable history	P0
COS-040	Synthetic follow-up mask/mesh geometry generation	P0

Expected deliverable: A measurable, rotatable tumor representation capable of showing authentic and simulated time states.

Phase 3 — Simulation and longitudinal intelligence

Workstream 06 — Synthetic longitudinal research engine

10 prompts

ID	Engineering prompt to create	Priority
COS-041	Synthetic simulation configuration and immutable branch data model	P0
COS-042	Baseline eligibility, clinical constraints and treatment compatibility engine	P0
COS-043	Strong tumor-response scenario generator	P0
COS-044	Partial-response and stable-tumor scenario generator	P0
COS-045	Progressive tumor-burden scenario generator	P0
COS-046	Initial-response-then-regrowth scenario generator	P0
COS-047	Clinical deterioration and treatment-toxicity scenario generator	P0
COS-048	Multimodal laboratory, vital, symptom and treatment-event generator	P0

COS-049	Scenario time scheduling, seeded variability and missingness modeling	P0
COS-050	Simulation validation, reproducibility, publication and failure isolation	P0

Expected deliverable: Five reproducible scenario families, with tumor evolution and patient health modeled independently where appropriate.

Workstream 07 — Temporal analysis and multimodal patient-state fusion

6 prompts

ID	Engineering prompt to create	Priority
COS-051	Patient observation timeline and temporal feature extraction	P0
COS-052	Deterministic tumor, laboratory and symptom trend calculations	P0
COS-053	Multimodal patient-state fusion and missingness/conflict handling	P0
COS-054	Longitudinal state update and dependent calculation orchestration	P0
COS-055	Baseline forecasting methods and model abstention behavior	P0
COS-056	Optional GRU or other temporal neural-network experimentation	P1

Expected deliverable: A longitudinal patient representation that produces valid comparisons without depending on an LLM.

Phase 4 — AI evidence and interaction

Workstream 08 — Clinical NLP, knowledge and AI interpretation

9 prompts

ID	Engineering prompt to create	Priority
COS-057	Meta Model API integration, provider adapter and structured output validation	P0
COS-058	Clinical note fact extraction, negation and temporal expression handling	P0
COS-059	Clinical concept and source-linked knowledge relationship model	P0

COS-060	Curated research evidence ingestion, licensing and versioned indexing	P0
COS-061	Evidence retrieval, relevance filtering and citation verification	P0
COS-062	Structured research interpretation context builder	P0
COS-063	Evidence-linked AI interpretation and explanatory output generation	P0
COS-064	Hallucination checks, numerical grounding, prohibited claims and abstention	P0
COS-065	AI result persistence, model audit, cache and outage fallback	P0

Expected deliverable: A grounded interpretation system that explains verified or simulated research observations and exposes uncertainty.

Workstream 09 — Backend APIs and service integration

6 prompts

ID	Engineering prompt to create	Priority
COS-066	Patient, diagnosis and observation API implementation	P0
COS-067	Imaging, segmentation and geometry API implementation	P0
COS-068	Simulation creation, branch management and status APIs	P0
COS-069	Digital twin state, timeline and comparison APIs	P0
COS-070	AI interpretation, evidence and evaluation APIs	P0
COS-071	Authentication, authorization, request validation and API contract testing	P0

Expected deliverable: A typed and secured API layer connecting all subsystems with consistent identifiers and documented error behavior.

Phase 5 — Frontend and demonstration experience

Workstream 10 — UI/UX and interactive digital twin dashboard

9 prompts

ID	Engineering prompt to create	Priority
COS-072	Global design system, navigation and responsive application shell	P0
COS-073	Research case dashboard and patient selection	P0
COS-074	Patient profile, source information and record completeness	P0
COS-075	Interactive 3D tumor viewer and camera controls	P0
COS-076	Imaging date selector, observed/synthetic state slider and tumor comparisons	P0
COS-077	Laboratory, vital-sign, symptom and treatment timeline charts	P0
COS-078	Simulation setup, generation progress and branch exploration	P0
COS-079	Evidence-linked AI insights, uncertainties and provenance presentation	P0
COS-080	Integrated UI states, accessibility, usability and failure handling	P0

Expected deliverable: A coherent clinician/researcher interface where the central digital twin is genuinely connected to backend data and changes.

Phase 6 — Evaluation, deployment and release

Workstream 11 — Scientific testing and model evaluation

7 prompts

ID	Engineering prompt to create	Priority
COS-081	Unit, schema, provenance and simulation integrity test suite	P0
COS-082	Imaging geometry, segmentation and physical measurement benchmark	P0
COS-083	Genuine longitudinal imaging and temporal baseline evaluation	P0
COS-084	Synthetic distribution and cross-modal plausibility evaluation	P0
COS-085	Clinical NLP extraction and evidence-retrieval evaluation	P0
COS-086	AI safety, hallucination, prompt-injection and abstention tests	P0

COS-087 End-to-end user journeys, regression tests and scientific evaluation report P0

Expected deliverable: Reproducible evidence of what the system works on, where it fails, and what is not clinically validated.

Workstream 12 — Infrastructure, security and deployment

6 prompts

ID	Engineering prompt to create	Priority
COS-088	Cloud deployment configuration, container images and environment separation	P0
COS-089	Object storage, artifact access and database backup policies	P0
COS-090	Inference-worker orchestration, resource limits and job processing	P0
COS-091	Security hardening, dataset permissions, secrets and external AI data controls	P0
COS-092	Logging, telemetry, health checks, error tracking and cost controls	P0
COS-093	Production-like deployment smoke tests and rollback procedure	P0

Expected deliverable: A deployed and recoverable application whose security, access controls and model usage are suitable for the authorized research demonstration.

Workstream 13 — Finalist demonstration and submission

5 prompts

ID	Engineering prompt to create	Priority
COS-094	Curate and verify five end-to-end synthetic scenario demonstration cases	P0
COS-095	Prepare authentic longitudinal imaging case and comparison demonstration	P0

COS-096	Implement deterministic demo fixtures, demo reset and graceful fallbacks	P0
COS-097	Prepare technical documentation, architecture visuals and evaluation evidence	P0
COS-098	Execute final presentation flow, deployment rehearsal and submission readiness audit	P0

Expected deliverable: A reproducible demonstration and submission package, not merely an application that works on one developer's laptop.

Phase 7 — System-wide final quality and integration reviews

These should be executed after major development milestones, particularly before the final release.

Workstream 14 — Final architecture and functional integrity audit

4 prompts

ID	Engineering prompt to create	Priority
COS-099	Full repository architecture and specification compliance audit	P0
COS-100	Cross-module integration, identity isolation and provenance audit	P0
COS-101	End-to-end functionality, dead controls and demonstration reliability audit	P0
COS-102	Risk-based remediation backlog, regression verification and release sign-off	P0

Workstream 15 — Final release operations

2 prompts

ID	Engineering prompt to create	Priority
COS-103	Final deployment verification, service recovery and backup validation	P0
COS-104	Freeze the validated release, tag the repository and assemble reproducibility evidence	P0

Recommended execution sequence

Although the inventory is numbered sequentially, the agent should not mechanically finish all tasks in one workstream before touching another. Some must be integrated early to reveal problems.

Implementation gates

Gate 1

Engineering governance

COS-001–015

A reproducible repository, consistent specifications and enforceable coding rules.

Gate 2

First authentic anatomical twin

COS-016–035, selected COS-066–075

One verified breast MRI becomes a correct measurable 3D lesion in the browser.

Gate 3

Longitudinal simulator

COS-036–054, COS-068–078

Five synthetic scenario families update tumor geometry and clinical timelines.

Gate 4

AI research intelligence

COS-055–065, COS-070 and COS-079

The system explains observed changes using validated data and traceable evidence.

Gate 5

Verification and deployment

COS-080–093

Independent tests, privacy checks and a recoverable deployment.

Gate 6

Finalist readiness

COS-094–104

Curated real and synthetic cases, full audits and final demonstration package.

The first prompts I would actually create

Start with COS-001 through COS-008, because they define the rules governing every subsequent code-writing task. In particular, COS-002 should audit all 12 specifications and produce a contradiction register. COS-003 should establish requirement-to-task traceability, and COS-007 should make tests mandatory before a coding agent reports completion.

After that, implement one small but complete vertical slice rather than generating dozens of disconnected modules:

Real MRI → validated mask → calculated tumor volume → browser 3D viewer → synthetic follow-up state → synchronized timeline → evidence-linked AI explanation.

Only then expand to the complete five-scenario engine, advanced functionality, and wider case library.

The master inventory intentionally avoids full genomic modeling, whole-body anatomy, autonomous therapy selection, drug discovery, and other systems excluded by the agreed hackathon scope.

One important constraint: Treat these 104 entries as an execution inventory, not 104 independent agent sessions that must all be completed separately. Closely related tasks may be combined into a bounded implementation session, while high-risk activities—medical data ingestion, spatial geometry, provenance, model evaluation, and security—should retain independent validation checkpoints.